20251117 Local Moran Like Correlation Coefficient

First i have to admit that the previous offset calcualtion is wrong (i.e., it was subtracted by the global mean values rather than the mean of each Σ_* bin), but now i have fixed it and our conclusion is still unchanged: such correlation or anticorrelation are still spatially varying. Here, I want to show how I calculate the local Moran-like correlation coefficient I of individual spaxel.

Moran-like product I_i

We start with two maps:

- $A(i)$ = $\Delta \log \Sigma_{\mathrm{SFR}}$ at spaxel i
- $B(i)$ = $\Delta[\mathrm{O/H}]$ at spaxel i

Take all valid spaxels and compute the mean and standard deviation in each Σ_* Bin:

$$\mu_A = \text{mean of } A, \quad \sigma_A = \text{std of } A \tag{1}$$

$$\mu_B = \text{mean of } B, \quad \sigma_B = \text{std of } B \tag{2}$$

Define z-scores:

$$z_A(i) = \frac{A(i) - \mu_A}{\sigma_A}, \qquad z_B(i) = \frac{B(i) - \mu_B}{\sigma_B}. \tag{3}$$

Here,

- $z_A(i) > 0$: SFR offset at i is above the galaxy mean (in units of σ).
- $z_A(i) < 0$: SFR offset at i is *below* the mean.

The same holds for metallicity: $z_B(i)$.

For each spaxel i, look at its neighbours j and average their z-values of B:

$$z_{\mathrm{lag,B}}(i) \equiv \frac{1}{N_i} \sum_{j \in N(i)} z_B(j), \tag{4}$$

where $N(i)$ is the set of neighbouring spaxels and N_i is its size, which is 8 here.

Here, $z_{\mathrm{lag,B}}(i)$ tells us whether the local metallicity environment around i is above or below the global mean (kind of spatial lag of property B at position i), in units of σ:

- $z_{\mathrm{lag,B}}(i) > 0$: neighbours of i tend to have high $\Delta[\mathrm{O/H}]$.
- $z_{\mathrm{lag,B}}(i) < 0$: neighbours of i tend to have low $\Delta[\mathrm{O/H}]$.

After this step, at each location we know:

- the value for the point itself: $z_A(i)$, and
- the value for its environment in B: $z_{\mathrm{lag,B}}(i)$.

Now we can define the Moran-like product:

$$I_i = z_A(i)\ z_{\mathrm{lag,B}}(i). \tag{5}$$

This product encodes agreement vs. disagreement between the point and its environment:

- If both are positive $(z_A(i) > 0$ and $z_{\mathrm{lag,B}}(i) > 0)$, then we have high $\Delta \log \Sigma_{\mathrm{SFR}}$ at i and a high-metallicity environment around it: High–High (HH), so $I_i > 0$.
- If both are negative $(z_A(i) < 0$ and $z_{\mathrm{lag,B}}(i) < 0)$, then we have low $\Delta \log \Sigma_{\mathrm{SFR}}$ at i and a low-metallicity environment: Low–Low (LL), again $I_i > 0$.
- If the signs are opposite $(z_A(i) > 0$, $z_{\mathrm{lag,B}}(i) < 0$ or vice versa), then we have a high-SFR offset surrounded by low metallicity, or a low-SFR offset surrounded by high metallicity: High–Low (HL) or Low–High (LH), yielding $I_i < 0$.

So:

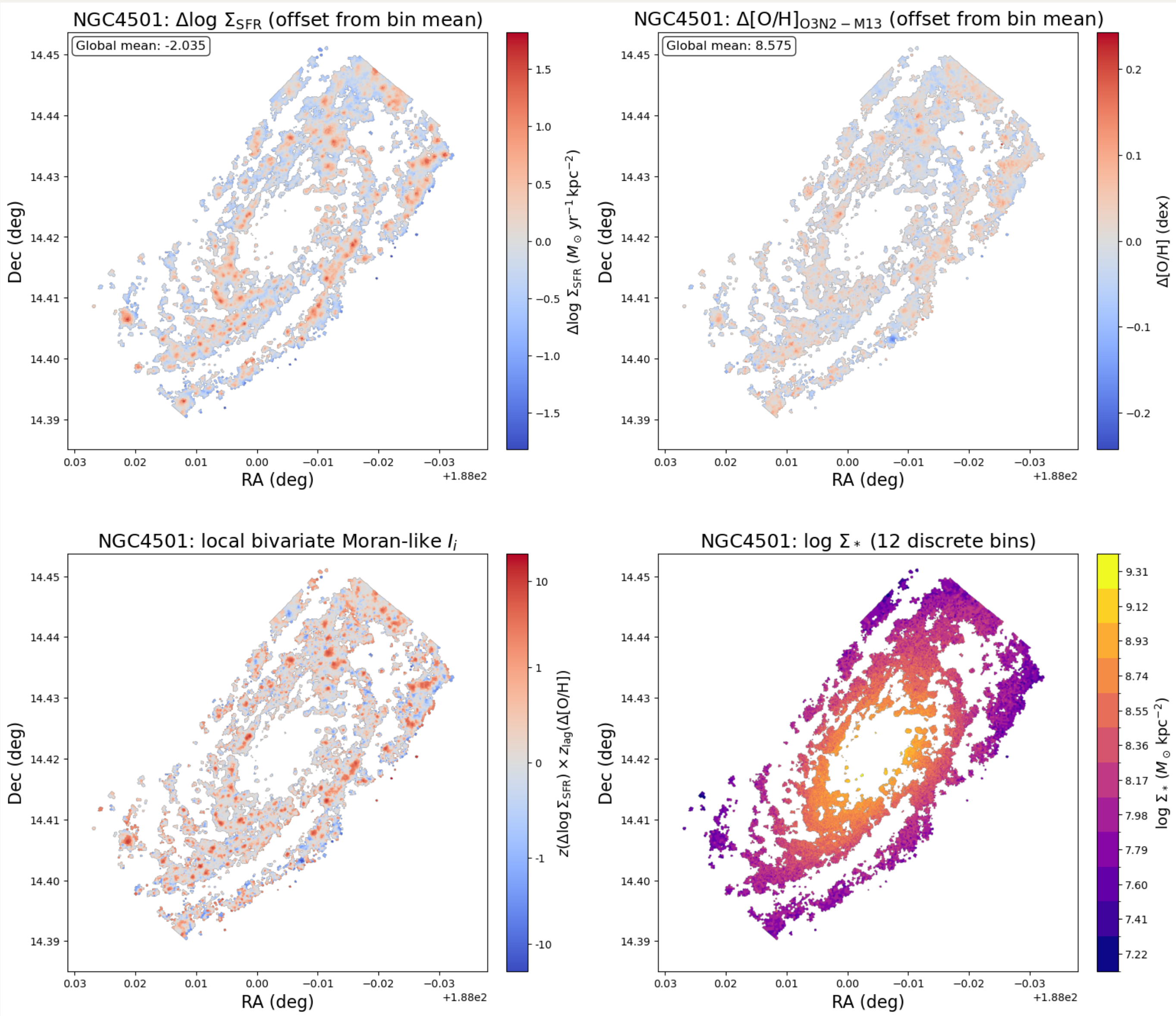
- The sign of I_i tells you whether the spaxel is *consistent* with its local metallicity environment (HH/LL) or *inverted* relative to it (HL/LH).

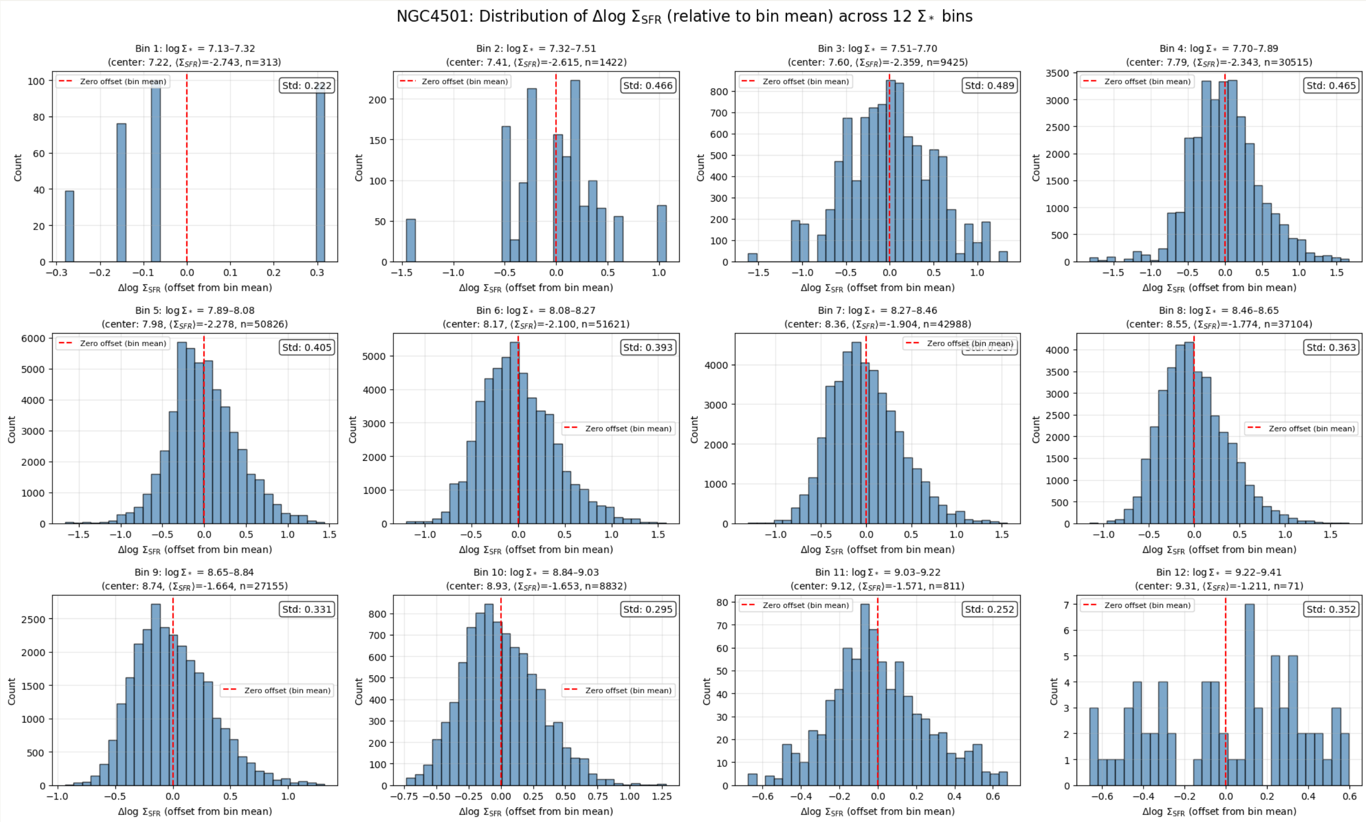
- The magnitude $|I_i|$ tells you how strong that pattern is.

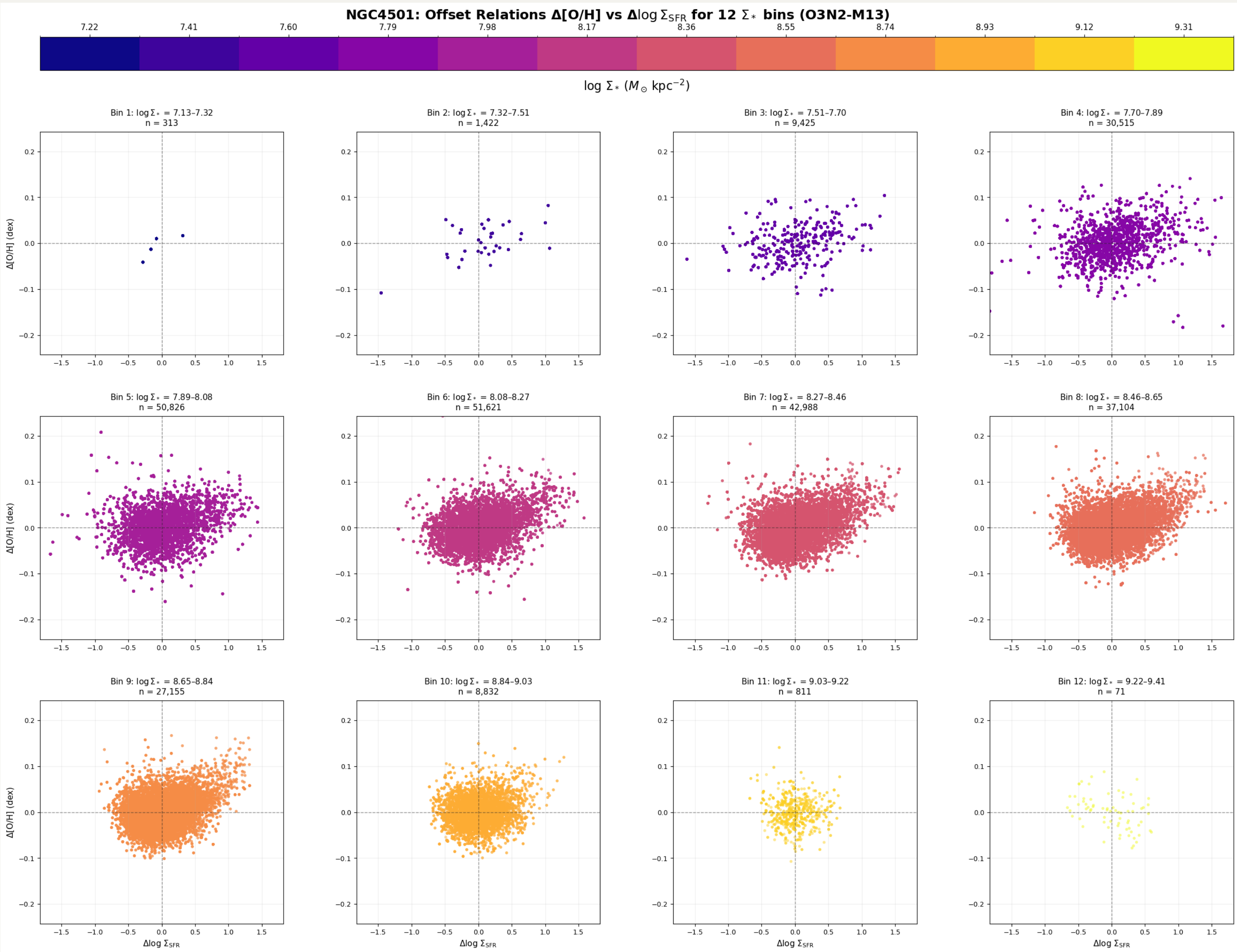
Therefore, I_i map shows where $\Delta \log \Sigma_{\rm SFR}$ is locally positively or negatively correlated with the surrounding $\Delta[{\rm O/H}]$ field.

Below, I will show 1) $\Delta \log \Sigma_{\rm SFR}$, $\Delta[{\rm O/H}]$, Moran-like product I_i and discrete Σ_* maps; 2) histograms of $\Delta \log(\Sigma_{\rm SFR})$ to check calculation of offset; and 3) offset relations in 12 Σ_* bins of NGC4501 (mostly positive correlation), NGC4396 (mostly negative correlation) and NGC4192 (seems to accross both low and high Σ_* regimes, so you will have a sense of the offset relations go from a bit decreasing trends to a bit increasing trends), respectively.

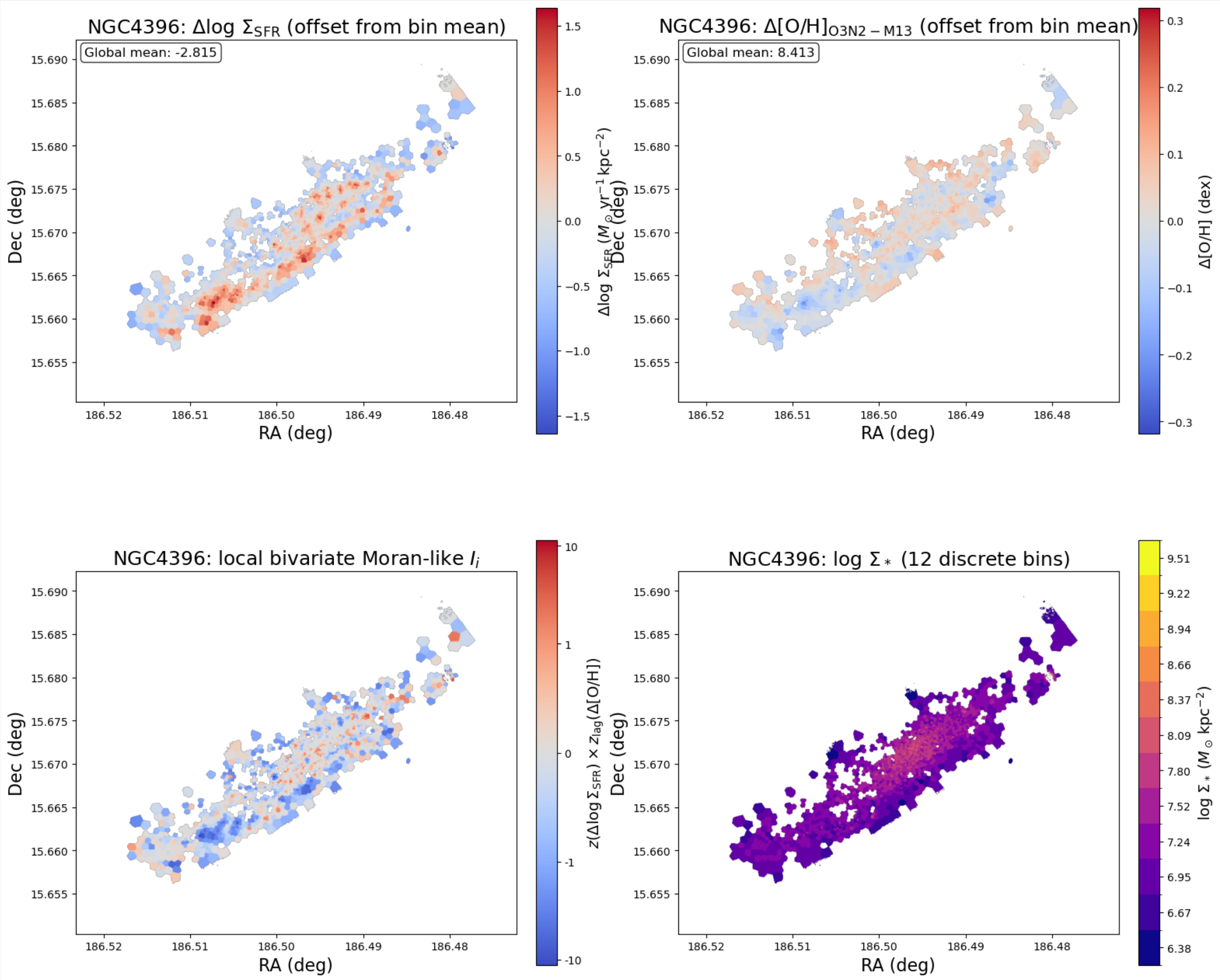
NGC4501



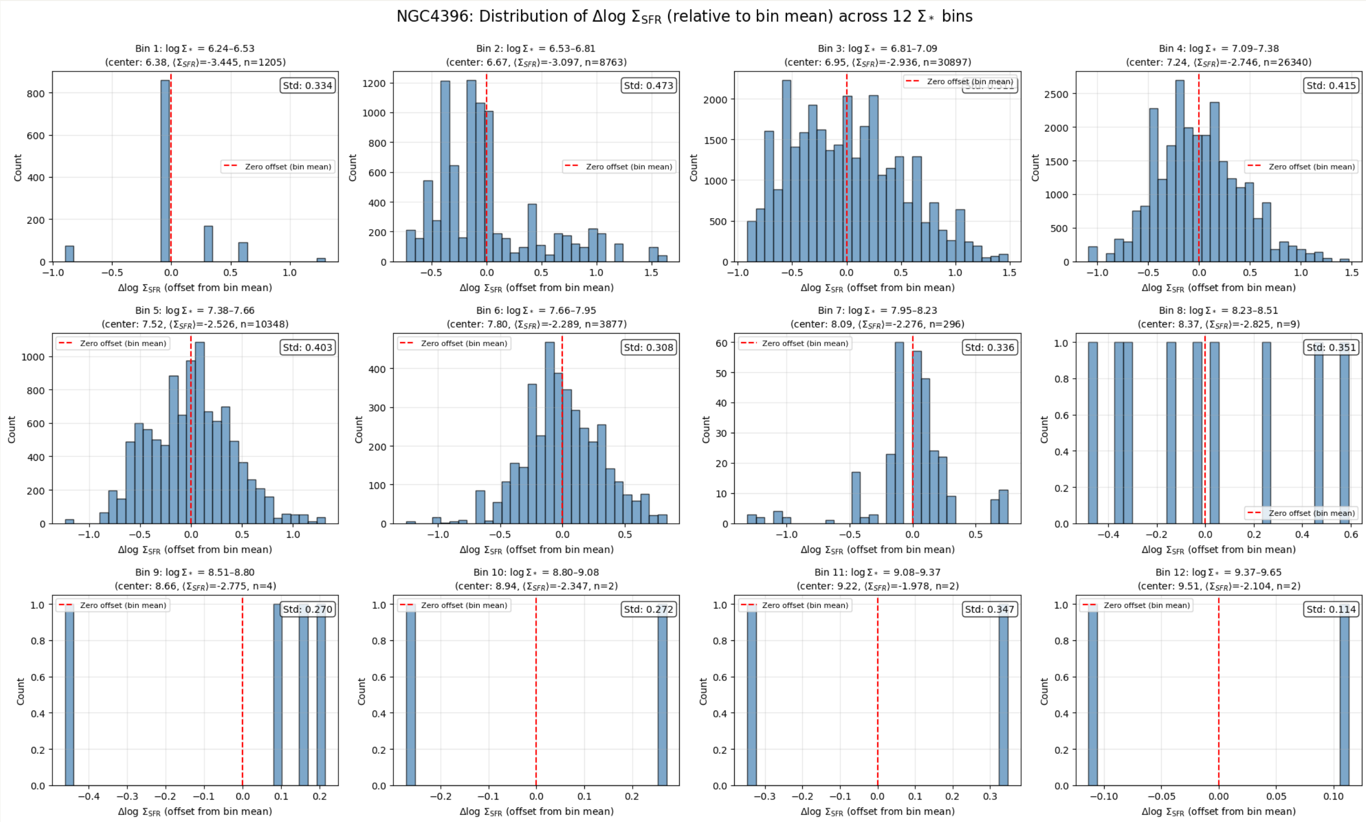

NGC4501: Distribution of Δlog Σ_SFR (relative to bin mean) across 12 Σ* bins
Bin 1: log Σ* = 7.13–7.32 (center: 7.22, ⟨Σ_SFR⟩=-2.743, n=313)
Std: 0.222
Bin 2: log Σ* = 7.32–7.51 (center: 7.41, ⟨Σ_SFR⟩=-2.615, n=1422)
Std: 0.466
Bin 3: log Σ* = 7.51–7.70 (center: 7.60, ⟨Σ_SFR⟩=-2.359, n=9425)
Std: 0.489
Bin 4: log Σ* = 7.70–7.89 (center: 7.79, ⟨Σ_SFR⟩=-2.343, n=30515)
Std: 0.465
Bin 5: log Σ* = 7.89–8.08 (center: 7.98, ⟨Σ_SFR⟩=-2.278, n=50826)
Std: 0.405
Bin 6: log Σ* = 8.08–8.27 (center: 8.17, ⟨Σ_SFR⟩=-2.100, n=51621)
Std: 0.393
Bin 7: log Σ* = 8.27–8.46 (center: 8.36, ⟨Σ_SFR⟩=-1.904, n=42988)
Bin 8: log Σ* = 8.46–8.65 (center: 8.55, ⟨Σ_SFR⟩=-1.774, n=37104)
Std: 0.363
Bin 9: log Σ* = 8.65–8.84 (center: 8.74, ⟨Σ_SFR⟩=-1.664, n=27155)
Std: 0.331
Bin 10: log Σ* = 8.84–9.03 (center: 8.93, ⟨Σ_SFR⟩=-1.653, n=8832)
Std: 0.295
Bin 11: log Σ* = 9.03–9.22 (center: 9.12, ⟨Σ_SFR⟩=-1.571, n=811)
Std: 0.252
Bin 12: log Σ* = 9.22–9.41 (center: 9.31, ⟨Σ_SFR⟩=-1.211, n=71)
Std: 0.352
Zero offset (bin mean)
Count
Δlog Σ_SFR (offset from bin mean)

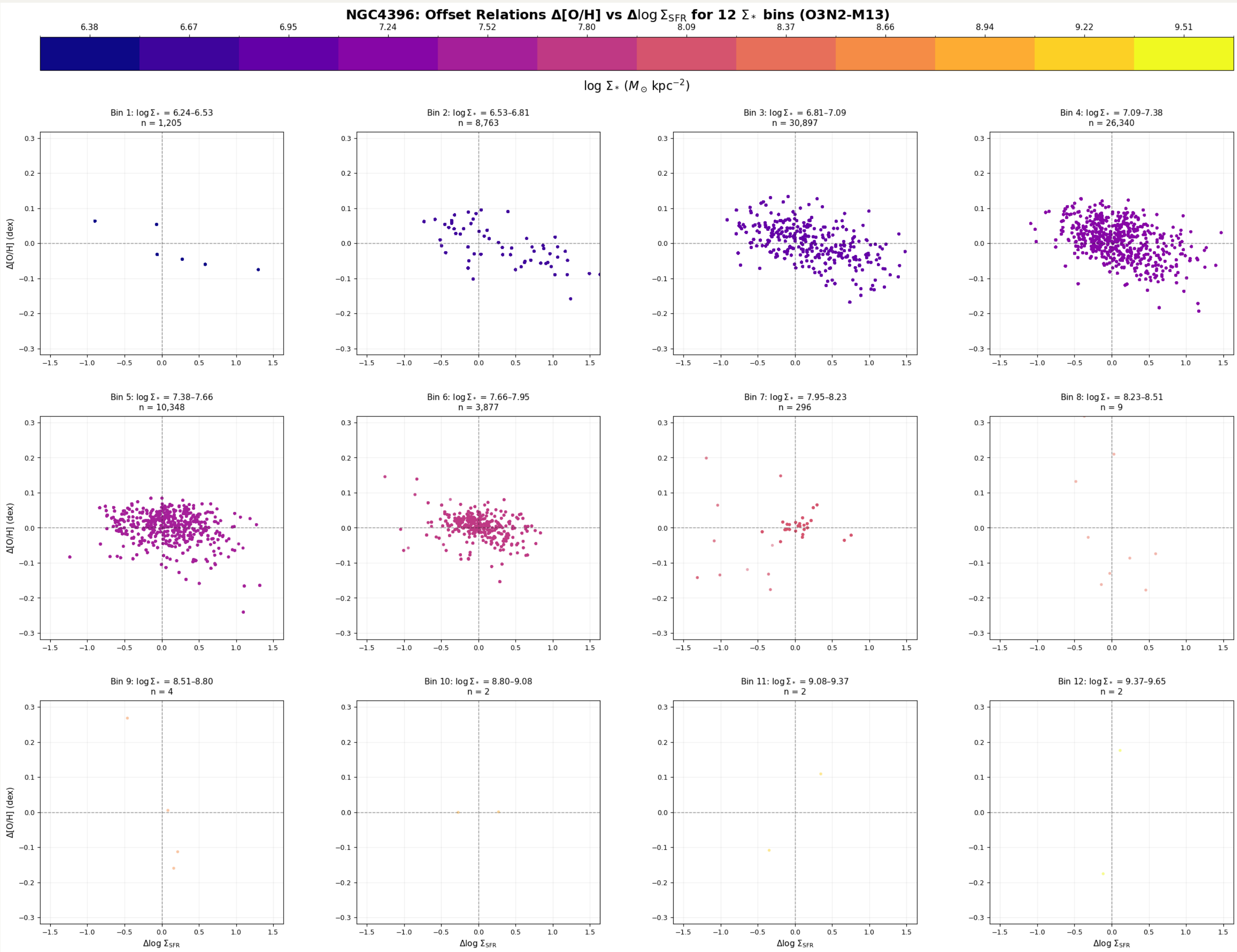

NGC4501: Offset Relations Δ[O/H] vs Δlog Σ_SFR for 12 Σ* bins (O3N2-M13)
7.22 7.41 7.60 7.79 7.98 8.17 8.36 8.55 8.74 8.93 9.12 9.31
log Σ* (M☉ kpc⁻²)
Bin 1: log Σ* = 7.13–7.32 n = 313
Bin 2: log Σ* = 7.32–7.51 n = 1,422
Bin 3: log Σ* = 7.51–7.70 n = 9,425
Bin 4: log Σ* = 7.70–7.89 n = 30,515
Bin 5: log Σ* = 7.89–8.08 n = 50,826
Bin 6: log Σ* = 8.08–8.27 n = 51,621
Bin 7: log Σ* = 8.27–8.46 n = 42,988
Bin 8: log Σ* = 8.46–8.65 n = 37,104
Bin 9: log Σ* = 8.65–8.84 n = 27,155
Bin 10: log Σ* = 8.84–9.03 n = 8,832
Bin 11: log Σ* = 9.03–9.22 n = 811
Bin 12: log Σ* = 9.22–9.41 n = 71
Δ[O/H] (dex)
Δlog Σ_SFR

NGC4396

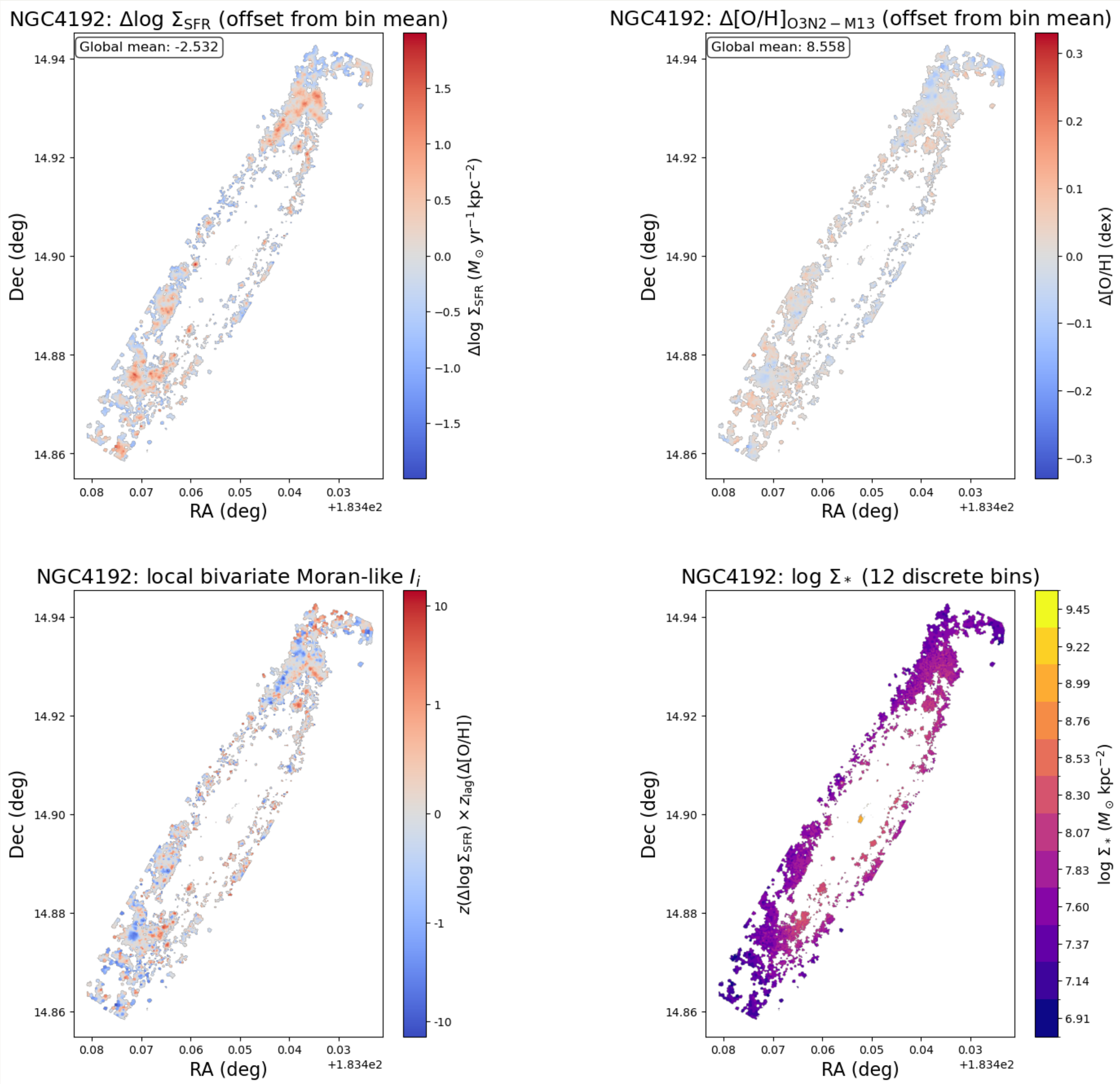


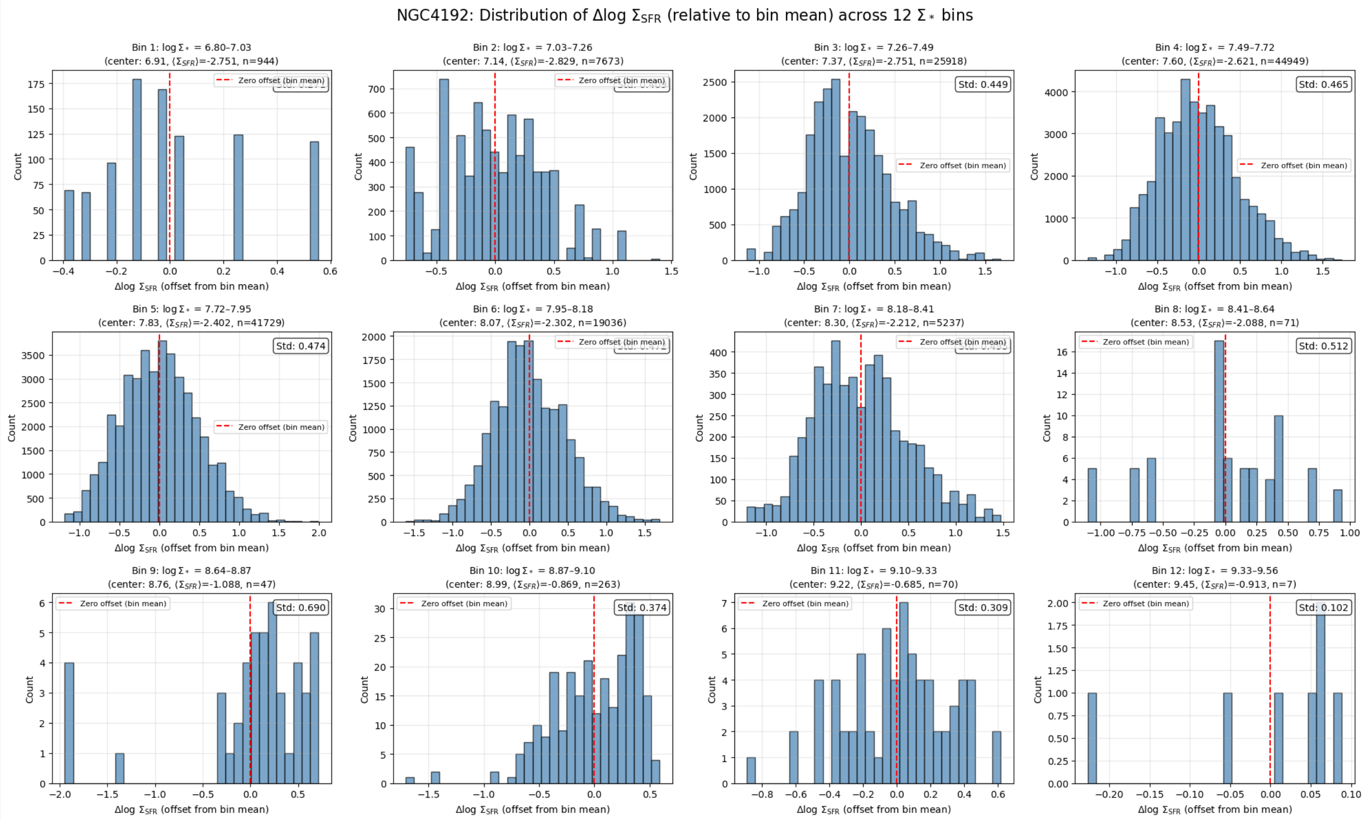
NGC4396: Δlog Σ_{SFR} (offset from bin mean)
Global mean: -2.815
NGC4396: Δ[O/H]$_{O3N2-M13}$ (offset from bin mean)
Global mean: 8.413
NGC4396: local bivariate Moran-like I_i
NGC4396: log Σ_* (12 discrete bins)
RA (deg)
Dec (deg)
Δlog Σ_{SFR} ($M_\odot$ yr^{-1} kpc^{-2})
Δ[O/H] (dex)
z(Δlog Σ_{SFR}) × z_{lag}(Δ[O/H])
log Σ_* ($M_\odot$ kpc^{-2})


NGC4396: Distribution of Δlog Σ_{SFR} (relative to bin mean) across 12 Σ_* bins
Bin 1: log Σ_* = 6.24–6.53 (center: 6.38, ⟨Σ_{SFR}⟩=-3.445, n=1205)
Bin 2: log Σ_* = 6.53–6.81 (center: 6.67, ⟨Σ_{SFR}⟩=-3.097, n=8763)
Bin 3: log Σ_* = 6.81–7.09 (center: 6.95, ⟨Σ_{SFR}⟩=-2.936, n=30897)
Bin 4: log Σ_* = 7.09–7.38 (center: 7.24, ⟨Σ_{SFR}⟩=-2.746, n=26340)
Bin 5: log Σ_* = 7.38–7.66 (center: 7.52, ⟨Σ_{SFR}⟩=-2.526, n=10348)
Bin 6: log Σ_* = 7.66–7.95 (center: 7.80, ⟨Σ_{SFR}⟩=-2.289, n=3877)
Bin 7: log Σ_* = 7.95–8.23 (center: 8.09, ⟨Σ_{SFR}⟩=-2.276, n=296)
Bin 8: log Σ_* = 8.23–8.51 (center: 8.37, ⟨Σ_{SFR}⟩=-2.825, n=9)
Bin 9: log Σ_* = 8.51–8.80 (center: 8.66, ⟨Σ_{SFR}⟩=-2.775, n=4)
Bin 10: log Σ_* = 8.80–9.08 (center: 8.94, ⟨Σ_{SFR}⟩=-2.347, n=2)
Bin 11: log Σ_* = 9.08–9.37 (center: 9.22, ⟨Σ_{SFR}⟩=-1.978, n=2)
Bin 12: log Σ_* = 9.37–9.65 (center: 9.51, ⟨Σ_{SFR}⟩=-2.104, n=2)
Std: 0.334
Std: 0.473
Std: 0.415
Std: 0.403
Std: 0.308
Std: 0.336
Std: 0.351
Std: 0.270
Std: 0.272
Std: 0.347
Std: 0.114
Zero offset (bin mean)
Count
Δlog Σ_{SFR} (offset from bin mean)

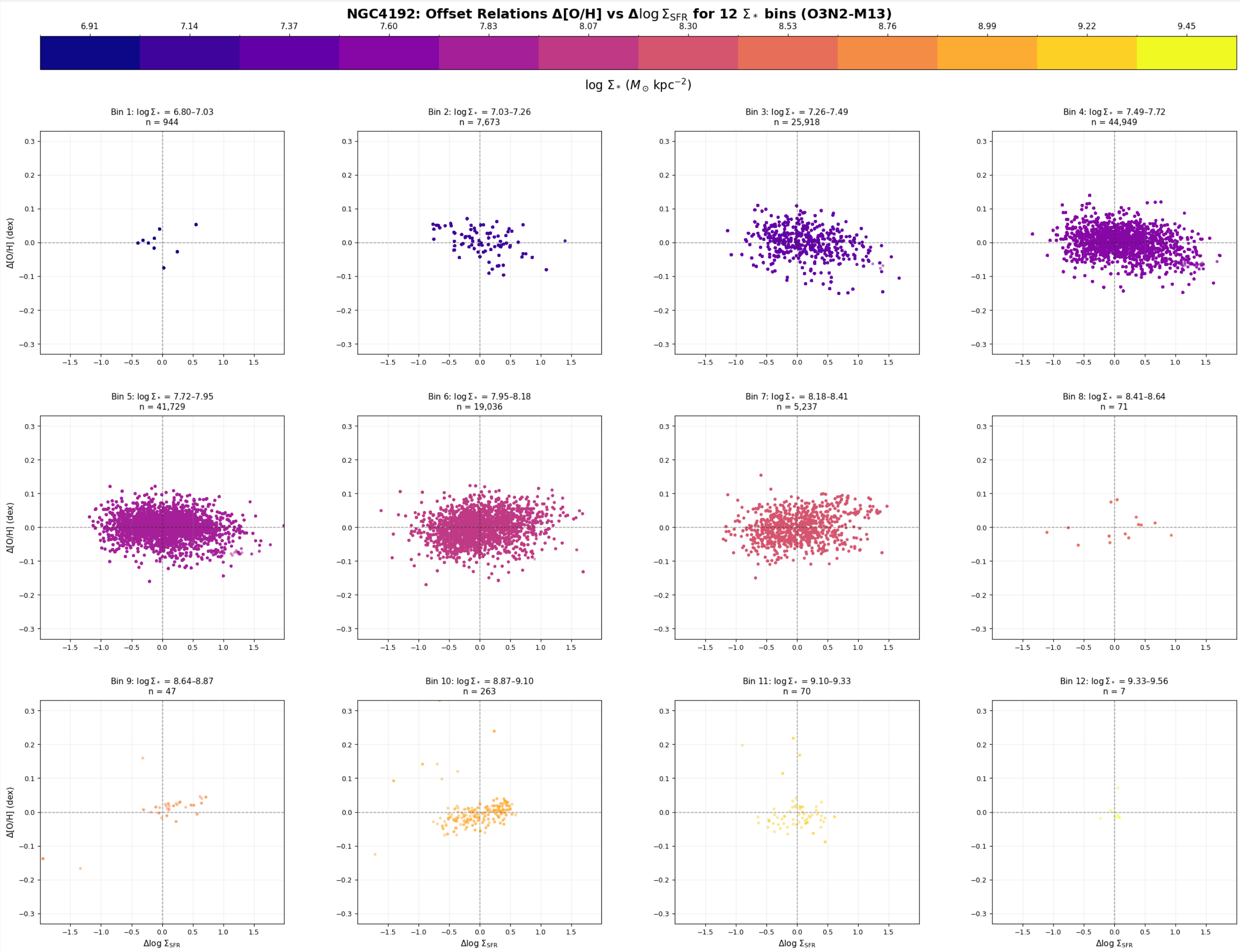

NGC4396: Offset Relations Δ[O/H] vs Δlog Σ_{SFR} for 12 Σ_* bins (O3N2-M13)
6.38
6.67
6.95
7.24
7.52
7.80
8.09
8.37
8.66
8.94
9.22
9.51
log Σ_* ($M_\odot$ kpc^{-2})
Bin 1: log Σ_* = 6.24–6.53 n = 1,205
Bin 2: log Σ_* = 6.53–6.81 n = 8,763
Bin 3: log Σ_* = 6.81–7.09 n = 30,897
Bin 4: log Σ_* = 7.09–7.38 n = 26,340
Bin 5: log Σ_* = 7.38–7.66 n = 10,348
Bin 6: log Σ_* = 7.66–7.95 n = 3,877
Bin 7: log Σ_* = 7.95–8.23 n = 296
Bin 8: log Σ_* = 8.23–8.51 n = 9
Bin 9: log Σ_* = 8.51–8.80 n = 4
Bin 10: log Σ_* = 8.80–9.08 n = 2
Bin 11: log Σ_* = 9.08–9.37 n = 2
Bin 12: log Σ_* = 9.37–9.65 n = 2
Δ[O/H] (dex)
Δlog Σ_{SFR}

NGC4192


NGC4192: Δlog Σ_SFR (offset from bin mean)
Global mean: -2.532
Δlog Σ_SFR (M_⊙ yr^-1 kpc^-2)
NGC4192: Δ[O/H]_O3N2−M13 (offset from bin mean)
Global mean: 8.558
Δ[O/H] (dex)
NGC4192: local bivariate Moran-like I_i
z(Δlog Σ_SFR) × z_lag(Δ[O/H])
NGC4192: log Σ_* (12 discrete bins)
log Σ_* (M_⊙ kpc^-2)
Dec (deg)
RA (deg)
+1.834e2



NGC4192: Distribution of Δlog Σ_{SFR} (relative to bin mean) across 12 Σ_* bins
Bin 1: log Σ_* = 6.80–7.03 (center: 6.91, ⟨Σ_{SFR}⟩=-2.751, n=944)
Bin 2: log Σ_* = 7.03–7.26 (center: 7.14, ⟨Σ_{SFR}⟩=-2.829, n=7673)
Bin 3: log Σ_* = 7.26–7.49 (center: 7.37, ⟨Σ_{SFR}⟩=-2.751, n=25918)
Std: 0.449
Bin 4: log Σ_* = 7.49–7.72 (center: 7.60, ⟨Σ_{SFR}⟩=-2.621, n=44949)
Std: 0.465
Bin 5: log Σ_* = 7.72–7.95 (center: 7.83, ⟨Σ_{SFR}⟩=-2.402, n=41729)
Std: 0.474
Bin 6: log Σ_* = 7.95–8.18 (center: 8.07, ⟨Σ_{SFR}⟩=-2.302, n=19036)
Bin 7: log Σ_* = 8.18–8.41 (center: 8.30, ⟨Σ_{SFR}⟩=-2.212, n=5237)
Bin 8: log Σ_* = 8.41–8.64 (center: 8.53, ⟨Σ_{SFR}⟩=-2.088, n=71)
Std: 0.512
Bin 9: log Σ_* = 8.64–8.87 (center: 8.76, ⟨Σ_{SFR}⟩=-1.088, n=47)
Std: 0.690
Bin 10: log Σ_* = 8.87–9.10 (center: 8.99, ⟨Σ_{SFR}⟩=-0.869, n=263)
Std: 0.374
Bin 11: log Σ_* = 9.10–9.33 (center: 9.22, ⟨Σ_{SFR}⟩=-0.685, n=70)
Std: 0.309
Bin 12: log Σ_* = 9.33–9.56 (center: 9.45, ⟨Σ_{SFR}⟩=-0.913, n=7)
Std: 0.102
Zero offset (bin mean)
Count
Δlog Σ_{SFR} (offset from bin mean)



NGC4192: Offset Relations Δ[O/H] vs Δlog Σ_{SFR} for 12 Σ_* bins (O3N2-M13)
6.91 7.14 7.37 7.60 7.83 8.07 8.30 8.53 8.76 8.99 9.22 9.45
log Σ_* ($M_\odot$ kpc^{-2})
Bin 1: log Σ_* = 6.80–7.03 n = 944
Bin 2: log Σ_* = 7.03–7.26 n = 7,673
Bin 3: log Σ_* = 7.26–7.49 n = 25,918
Bin 4: log Σ_* = 7.49–7.72 n = 44,949
Bin 5: log Σ_* = 7.72–7.95 n = 41,729
Bin 6: log Σ_* = 7.95–8.18 n = 19,036
Bin 7: log Σ_* = 8.18–8.41 n = 5,237
Bin 8: log Σ_* = 8.41–8.64 n = 71
Bin 9: log Σ_* = 8.64–8.87 n = 47
Bin 10: log Σ_* = 8.87–9.10 n = 263
Bin 11: log Σ_* = 9.10–9.33 n = 70
Bin 12: log Σ_* = 9.33–9.56 n = 7
Δ[O/H] (dex)
Δlog Σ_{SFR}